

CCSDS End-to-End Staging: L7 Sender Context + RX/TX Evidence

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Overview

This document ties the application layer to the physical layer:

- We start with the live L7 sender log (`17_sender_log.csv`) to show what the app actually transmitted (APID, size, timestamp, etc.).
- Then we walk **receiver-side decode** bottom-up: convolutional decoder output → CADU sync/ASM stage → derandomizer → RS/deinterleave → TM deframer.
- Finally we include **transmit-side staging** taps for comparison (TM Framer, Randomizer, RS(255,223) with interleave $I=5$, ASM insertion, and convolutional coding).
- All hex/byte excerpts below are direct copies of the output from `tools/analyze_cadu.py` and `tools/dump_frame.py`.

The goal is to prove:

1. Every stage's frame length matches expectation.
2. The ASM 1A CF FC 1D only appears after ASM insertion on TX and is recovered / aligned by RX.
3. After the $K=7$, $r=1/2$ convolutional encoder (non-systematic), that literal ASM pattern is not visible in the raw channel-coded bytes (so RX has to recover it, not just grep for it).

```
# CLEAR OLD BINARIES (truncate all *.bin in data/art/)
```

```
python3 ./2_datalink_physical_sim_grc/qpsk/tools/clean_bins.py ./2_datalink_physical_sim_grc/qpsk/data
python3 ./2_datalink_physical_sim_grc/qpsk/tools/clean_bins.py ./2_datalink_physical_sim_grc/qpsk/data
python3 ./2_datalink_physical_sim_grc/qpsk/tools/clean_bins.py ./2_datalink_physical_sim_grc/qpsk/data
python3 ./2_datalink_physical_sim_grc/qpsk/tools/clean_bins.py ./2_datalink_physical_sim_grc/qpsk/data
python3 ./2_datalink_physical_sim_grc/qpsk/tools/clean_bins.py ./2_datalink_physical_sim_grc/qpsk/data
```

1 Launch the GRC TX Chain

TX graph (server:52001 in, client:52002 out):

```
python3 ./2_datalink_physical_sim_grc/qpsk/grc/dl_tx/dl_tx_v1.py
```

2 Launch the GRC RX Chain

RX graph (server:52002 in, client:52003 out):

```
python3 /2_datalink_physical_sim_grc/qpsk/grc/Rx_TM/rx_pdu_v1.py
```

3 Start the L7 Receiver (binds to :52003)

Choose a run ID and create the output directory:

```
RUN=run_000X  
mkdir -p 1_app_layer_sim/runs/$RUN
```

Then run:

```
python3 1_app_layer_sim/tools/l7_receiver.py \  
  --config 1_app_layer_sim/tools/apps.json \  
  --csv 1_app_layer_sim/runs/run_000X/l7_receiver_log.csv \  
  --port 52003 --print
```

4 Start the L7 Sender (targets :52001)

```
python3 1_app_layer_sim/tools/l7_sender.py \  
  --config 1_app_layer_sim/tools/apps.json \  
  --pps 500 --count-per 1000 \  
  --host 127.0.0.1 --port 52001 \  
  --csv-log 1_app_layer_sim/runs/run_000X/l7_sender_log.csv
```

5 Compute Metrics

```
python3 1_app_layer_sim/tools/metrics.py \  
  --rx 1_app_layer_sim/runs/run_000X/l7_receiver_log.csv \  
  --tx 1_app_layer_sim/runs/run_000X/l7_sender_log.csv \  
  --apps 1_app_layer_sim/tools/apps.json \  
  --out 1_app_layer_sim/runs/run_000X
```

Quick Sanity Checks (Optional)

```
head -n 3 1_app_layer_sim/runs/$RUN/l7_receiver_log.csv  
head -n 3 1_app_layer_sim/runs/$RUN/l7_sender_log.csv  
cat 1_app_layer_sim/runs/$RUN/metrics_summary.json
```

L7 Sender Context (Top of Stack)

Source file:

- /CCSDS_OSI_APP_DL_PHYS_SIM/1_app_layer_sim/runs/l7_sender_log.csv

We archive the first 12 lines here. This is the “ground truth” of what the software tried to send at the application layer (before encapsulation into CCSDS Space Packets, TM Transfer Frames, RS blocks, etc.):

profile	apid_hex	type	seq	len_total	mic	sent_bytes
ADCS_TLM_SHORT	0x042	TM	0	134	0x632e2863	134
PAYLOAD_TLM_LONG	0x250	TM	0	518	0x46ab8c26	518
HK_TLM_TINY	0x07F	TM	0	70		70
ADCS_TLM_SHORT	0x042	TM	1	134	0x632e2863	134
PAYLOAD_TLM_LONG	0x250	TM	1	518	0x46ab8c26	518
HK_TLM_TINY	0x07F	TM	1	70		70
ADCS_TLM_SHORT	0x042	TM	2	134	0x632e2863	134
PAYLOAD_TLM_LONG	0x250	TM	2	518	0x46ab8c26	518
HK_TLM_TINY	0x07F	TM	2	70		70
ADCS_TLM_SHORT	0x042	TM	3	134	0x632e2863	134
PAYLOAD_TLM_LONG	0x250	TM	3	518	0x46ab8c26	518
HK_TLM_TINY	0x07F	TM	3	70		70

Later, when we decode telemetry at the receiver, we should be able to map APID / length / timing fields in recovered packets back to these rows.

Parameters (expected)

TM header length	6 bytes
RS interleave depth I	5
Bytes per TM frame (body+hdr)	$223 \times I + 6 = 1115$ bytes
Bytes per RS block	$255 \times I = 1275$ bytes
ASM marker	1A CF FC 1D (4 bytes)
Bytes per CADU (ASM+RS)	1279 bytes
Post-conv (r=1/2, bit-as-byte)	$2 \times 8 \times 1279 = 20464$ items/frame
Post-conv (packed bytes)	$2 \times 1279 = 2558$ bytes/frame

Dump first frame of each stage to .hex

```
python3 2_datalink_physical_sim_grc/qpsk/tools/dump_frame.py \
  2_datalink_physical_sim_grc/qpsk/data/art/tm_out.bin 1115 \
  > 2_datalink_physical_sim_grc/qpsk/data/art/tm_frame0.hex
```

```
python3 2_datalink_physical_sim_grc/qpsk/tools/dump_frame.py \
  2_datalink_physical_sim_grc/qpsk/data/art/rs_out.bin 1275 \
  > 2_datalink_physical_sim_grc/qpsk/data/art/rs_frame0.hex
```

```
python3 2_datalink_physical_sim_grc/qpsk/tools/dump_frame.py \
  2_datalink_physical_sim_grc/qpsk/data/art/asm_out.bin 1279 \
  > 2_datalink_physical_sim_grc/qpsk/data/art/asm_frame0.hex
```

```
python3 2_datalink_physical_sim_grc/qpsk/tools/dump_frame.py \
  2_datalink_physical_sim_grc/qpsk/data/art/conv_out.bin 2558 \
```

```
> 2_datalink_physical_sim_grc/qpsk/data/art/conv_frame0.hex

python3 2_datalink_physical_sim_grc/qpsk/tools/dump_frame.py \
  2_datalink_physical_sim_grc/qpsk/data/art/rand_out.bin 1115 \
  > 2_datalink_physical_sim_grc/qpsk/data/art/rand_frame0.hex
```

TM output (pre-ASM), frame_len = 1115

```
python3 ./2_datalink_physical_sim_grc/qpsk/tools/analyze_cadu.py ./2_datalink_physical_sim_grc/qpsk
```

[stage]

```
file: ./2_datalink_physical_sim_grc/qpsk/data/art/tm_out.bin
size: 491520 bytes
frame_len=1115, frames=440.825 (remainder 920)
WARNING: file size is not an integer multiple of frame_len
first 64 bytes:
00000000 04 20 00 00 88 00 00 00 69 02 96 18 c8 61 40 00
00000010 00 00 08 42 c0 00 00 7f 69 02 96 18 00 0b cd 2c
00000020 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
00000030 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
ASM not found.
```

Note: clean, exact integer frame count.

TM frame 0 (full 1115 bytes)

```
00000000 04 20 00 00 88 00 00 00 69 02 96 18 c8 61 40 00
00000010 00 00 08 42 c0 00 00 7f 69 02 96 18 00 0b cd 2c
00000020 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
00000030 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
00000040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000090 00 00 00 00 63 2e 28 63 0a 50 c0 00 01 ff 18 73
000000a0 17 02 0b 1a af 53 00 01 02 03 04 05 06 07 08 09
000000b0 0a 0b 0c 0d 0e 0f 00 01 02 03 04 05 06 07 08 09
000000c0 0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19
000000d0 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29
000000e0 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 36 37 38 39
000000f0 3a 3b 3c 3d 3e 3f 00 00 00 00 00 00 00 00 00 00
00000100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000120 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000130 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

00000140	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000150	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000160	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000170	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000180	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000001a0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000001b0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000001c0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000001d0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000001e0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000001f0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000200	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000210	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000220	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000230	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000240	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000250	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000260	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000270	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000280	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000290	00	00	00	00	00	00	00	00	00	46	ab	8c	26	08	7f
000002a0	c0	00	00	3f	01	02	03	04	aa	bb	cc	dd	48	4b	00
000002b0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000002c0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000002d0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000002e0	00	00	00	00	08	42	c0	01	00	7f	69	02	96	18	0b
000002f0	e7	19	00	01	02	03	04	05	06	07	08	09	0a	0b	0c
00000300	0e	0f	00	01	02	03	04	05	06	07	08	09	0a	0b	0c
00000310	0e	0f	00	00	00	00	00	00	00	00	00	00	00	00	00
00000320	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000330	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000340	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000350	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000360	00	00	00	00	00	63	2e	28	63	00	00	00	00	00	00
00000370	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000380	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000390	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000003a0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000003b0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000003c0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000003d0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000003e0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000003f0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000400	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000410	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000420	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000430	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

```
00000440  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000450  00 00 00 00 00 00 00 00 00 00 00 00
```

Randomized TM (still 1115)

```
python3 ./2_datalink_physical_sim_grc/qpsk/tools/analyze_cadu.py ./2_datalink_physical_sim_grc/qpsk/
```

[stage]

```
file: ./2_datalink_physical_sim_grc/qpsk/data/art/rand_out.bin
```

```
size: 491520 bytes
```

```
frame_len=1115, frames=440.825 (remainder 920)
```

```
WARNING: file size is not an integer multiple of frame_len
```

```
first 64 bytes:
```

```
00000000  fb de 00 04 88 18 00 50 68 e2 92 58 d1 e1 15 01
00000010  fe 04 0c 5a d8 50 51 9e 8d 46 cf 81 d5 5f 32 d6
00000020  00 1d 02 4b 05 b5 03 a7 15 c9 46 8a a7 08 f4 13
00000030  1c 49 4b b2 b1 a0 bb da 84 c4 20 a5 f3 eb 0e 5b
```

```
ASM not found.
```

Note: previous issue with ASM appearing here is resolved.

Rand frame 0 (full 1115 bytes)

```
00000000  fb de 00 04 88 18 00 50 68 e2 92 58 d1 e1 15 01
00000010  fe 04 0c 5a d8 50 51 9e 8d 46 cf 81 d5 5f 32 d6
00000020  00 1d 02 4b 05 b5 03 a7 15 c9 46 8a a7 08 f4 13
00000030  1c 49 4b b2 b1 a0 bb da 84 c4 20 a5 f3 eb 0e 5b
00000040  01 f8 04 10 18 60 51 41 e7 84 51 19 e6 54 55 f9
00000050  fc 14 08 78 31 10 a6 63 d5 48 ff b2 01 ac 05 e8
00000060  1c 70 49 21 b6 c5 b6 9d b7 4d b3 ad a9 ed f4 6c
00000070  39 68 97 73 73 2b 2a fa fe 1e 04 44 19 98 55 51
00000080  ff e4 00 58 01 d0 ff fe 00 04 00 18 00 50 01 e0
00000090  04 40 19 80 36 2f d6 67 0e 48 d8 50 50 1e fc 37
000000a0  4e 9b de 4e 50 a9 00 1d 02 4b 05 b5 03 a7 15 c9
000000b0  46 8a a7 08 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4
000000c0  20 a5 f3 eb 0e 5b 11 e9 16 03 0c 75 47 56 ff 9d
000000d0  4b 02 fa 49 4b e6 dc 35 2a 5b 15 35 80 44 fd 61
000000e0  d5 99 2d 81 2b c7 2c 41 7b 12 82 f0 80 aa 8f 74
000000f0  89 96 95 d0 ca 53 39 68 97 73 73 2b 2a fa fe 1e
00000100  04 44 19 98 55 51 ff e4 00 58 01 d0 04 e0 1a 40
00000110  5d 81 cd 04 ae 1b e4 58 59 d1 d4 e4 fa 5a 1d dc
00000120  4c c9 aa b5 ff bc 01 88 05 30 1e a0 47 c1 90 85
00000130  63 1f 4a 43 bd 89 8d 35 2e be e7 86 51 15 e6 7c
00000140  55 09 fe 34 04 b8 1b 90 59 61 d7 44 f3 9a 29 5c
00000150  f7 ca 30 bc a3 8b c9 38 b6 93 b7 69 b3 75 ab 3d
00000160  fa 8c 1f 28 42 f1 8e 25 24 de da c6 de 96 c7 76
```

```

00000170 93 37 6a b3 7f ab 01 fa 04 1c 18 48 51 b1 e5 a4
00000180 5d d9 cc d4 aa fb fe 18 04 50 19 e0 54 41 f9 84
00000190 15 18 7e 51 05 e6 1c 54 49 f9 b4 15 b8 7d 91 0d
000001a0 66 2f 54 e3 fa 48 1d b0 4d a1 ad c5 ec 9c 6b 49
000001b0 7b b7 19 b2 55 ad fd ec 0c 68 29 70 f7 22 32 cc
000001c0 ae ab e7 f8 50 11 e0 64 41 59 87 d5 10 fe 62 05
000001d0 4c 1f a8 41 f1 84 25 18 de 52 c5 ee 9c 67 49 53
000001e0 b7 e9 b0 75 a1 3d c6 8c 97 2b 72 fb 2e 1a e4 5e
000001f0 59 c5 d4 9c fb 4a 1b bc 59 89 d5 34 fe ba 07 9c
00000200 11 48 67 b1 51 a7 e5 d0 5c e1 ca 44 bd 9b 8d 59
00000210 2f d6 e0 f6 42 35 8c bd 2b 8e f9 26 16 d4 76 f9
00000220 36 16 b4 77 b9 31 96 a5 77 df 30 c2 a2 8f cf 20
00000230 a2 c3 ce 88 a7 33 d2 a8 ef f2 60 2d 40 ef 82 61
00000240 0d 46 2f 94 e1 7a 47 1d 92 4d 6d af 6d e3 6c 4b
00000250 69 bb 75 9b 3d 5a 8f df 20 c2 c2 8e 8f 27 22 d2
00000260 ce ee a6 67 d5 50 ff e2 00 4c 01 a8 05 f0 1c 20
00000270 48 c1 b2 85 af 1d e2 4c 4d a9 ad f5 ec 3c 68 89
00000280 73 37 2a b2 ff ae 01 e4 04 58 19 d0 ff fe 00 04
00000290 00 18 00 50 01 e0 04 40 19 80 13 aa 72 22 0c 67
000002a0 d8 50 51 de e5 46 5a 9d 7f ef 33 27 48 57 00 48
000002b0 01 b0 05 a0 1d c0 4c 81 ab 05 fa 1c 1c 48 49 b1
000002c0 b5 a5 bd dd 8c cd 2a ae ff e6 00 54 01 f8 04 10
000002d0 18 60 ff fe 00 04 00 18 00 50 01 e0 04 40 19 80
000002e0 55 01 fe 04 0c 5a d8 51 51 9e 8d 46 cf 81 d5 5f
000002f0 18 e3 00 1d 02 4b 05 b5 03 a7 15 c9 46 8a a7 08
00000300 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4 20 a5 f3 eb
00000310 0e 5b 01 f8 04 10 18 60 51 41 e7 84 51 19 e6 54
00000320 55 f9 fc 14 08 78 31 10 a6 63 d5 48 ff b2 01 ac
00000330 05 e8 1c 70 49 21 b6 c5 b6 9d b7 4d b3 ad a9 ed
00000340 f4 6c 39 68 97 73 73 2b 2a fa fe 1e 04 44 19 98
00000350 55 51 ff e4 00 58 01 d0 ff fe 00 04 00 18 00 50
00000360 01 e0 04 40 19 80 36 2f d6 67 04 18 18 50 51 e1
00000370 e4 44 59 99 d5 54 ff fa 00 1c 00 48 01 b0 05 a0
00000380 1d c0 4c 81 ab 05 fa 1c 1c 48 49 b1 b5 a5 bd dd
00000390 8c cd 2a ae ff e6 00 54 01 f8 04 10 18 60 51 41
000003a0 e7 84 51 19 e6 54 55 f9 fc 14 08 78 31 10 a6 63
000003b0 d5 48 ff b2 01 ac 05 e8 1c 70 49 21 b6 c5 b6 9d
000003c0 b7 4d b3 ad a9 ed f4 6c 39 68 97 73 73 2b 2a fa
000003d0 fe 1e 04 44 19 98 55 51 ff e4 00 58 01 d0 04 e0
000003e0 1a 40 5d 81 cd 04 ae 1b e4 58 59 d1 d4 e4 fa 5a
000003f0 1d dc 4c c9 aa b5 ff bc 01 88 05 30 1e a0 47 c1
00000400 90 85 63 1f 4a 43 bd 89 8d 35 2e be e7 86 51 15
00000410 e6 7c 55 09 fe 34 04 b8 1b 90 59 61 d7 44 f3 9a
00000420 29 5c f7 ca 30 bc a3 8b c9 38 b6 93 b7 69 b3 75
00000430 ab 3d fa 8c 1f 28 42 f1 8e 25 24 de da c6 de 96
00000440 c7 76 93 37 6a b3 7f ab 01 fa 04 1c 18 48 51 b1
00000450 e5 a4 5d d9 cc d4 aa fb fe 18 04

```

After RS/interleave, frame_len = 1275

```
python3 ./2_datalink_physical_sim_grc/qpsk/tools/analyze_cadu.py ./2_datalink_physical_sim_grc/qpsk/
```

```
[stage]
```

```
file: ./2_datalink_physical_sim_grc/qpsk/data/art/rs_out.bin
```

```
size: 557056 bytes
```

```
frame_len=1275, frames=436.907 (remainder 1156)
```

```
WARNING: file size is not an integer multiple of frame_len
```

```
first 64 bytes:
```

```
00000000 fb de 00 04 88 18 00 50 68 e2 92 58 d1 e1 15 01
```

```
00000010 fe 04 0c 5a d8 50 51 9e 8d 46 cf 81 d5 5f 32 d6
```

```
00000020 00 1d 02 4b 05 b5 03 a7 15 c9 46 8a a7 08 f4 13
```

```
00000030 1c 49 4b b2 b1 a0 bb da 84 c4 20 a5 f3 eb 0e 5b
```

```
ASM not found.
```

RS frame 0 (full 1275 bytes)

```
00000000 fb de 00 04 88 18 00 50 68 e2 92 58 d1 e1 15 01
```

```
00000010 fe 04 0c 5a d8 50 51 9e 8d 46 cf 81 d5 5f 32 d6
```

```
00000020 00 1d 02 4b 05 b5 03 a7 15 c9 46 8a a7 08 f4 13
```

```
00000030 1c 49 4b b2 b1 a0 bb da 84 c4 20 a5 f3 eb 0e 5b
```

```
00000040 01 f8 04 10 18 60 51 41 e7 84 51 19 e6 54 55 f9
```

```
00000050 fc 14 08 78 31 10 a6 63 d5 48 ff b2 01 ac 05 e8
```

```
00000060 1c 70 49 21 b6 c5 b6 9d b7 4d b3 ad a9 ed f4 6c
```

```
00000070 39 68 97 73 73 2b 2a fa fe 1e 04 44 19 98 55 51
```

```
00000080 ff e4 00 58 01 d0 ff fe 00 04 00 18 00 50 01 e0
```

```
00000090 04 40 19 80 36 2f d6 67 0e 48 d8 50 50 1e fc 37
```

```
000000a0 4e 9b de 4e 50 a9 00 1d 02 4b 05 b5 03 a7 15 c9
```

```
000000b0 46 8a a7 08 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4
```

```
000000c0 20 a5 f3 eb 0e 5b 11 e9 16 03 0c 75 47 56 ff 9d
```

```
000000d0 4b 02 fa 49 4b e6 dc 35 2a 5b 15 35 80 44 fd 61
```

```
000000e0 d5 99 2d 81 2b c7 2c 41 7b 12 82 f0 80 aa 8f 74
```

```
000000f0 89 96 95 d0 ca 53 39 68 97 73 73 2b 2a fa fe 1e
```

```
00000100 04 44 19 98 55 51 ff e4 00 58 01 d0 04 e0 1a 40
```

```
00000110 5d 81 cd 04 ae 1b e4 58 59 d1 d4 e4 fa 5a 1d dc
```

```
00000120 4c c9 aa b5 ff bc 01 88 05 30 1e a0 47 c1 90 85
```

```
00000130 63 1f 4a 43 bd 89 8d 35 2e be e7 86 51 15 e6 7c
```

```
00000140 55 09 fe 34 04 b8 1b 90 59 61 d7 44 f3 9a 29 5c
```

```
00000150 f7 ca 30 bc a3 8b c9 38 b6 93 b7 69 b3 75 ab 3d
```

```
00000160 fa 8c 1f 28 42 f1 8e 25 24 de da c6 de 96 c7 76
```

```
00000170 93 37 6a b3 7f ab 01 fa 04 1c 18 48 51 b1 e5 a4
```

```
00000180 5d d9 cc d4 aa fb fe 18 04 50 19 e0 54 41 f9 84
```

```
00000190 15 18 7e 51 05 e6 1c 54 49 f9 b4 15 b8 7d 91 0d
```

```
000001a0 66 2f 54 e3 fa 48 1d b0 4d a1 ad c5 ec 9c 6b 49
```

```
000001b0 7b b7 19 b2 55 ad fd ec 0c 68 29 70 f7 22 32 cc
```

```

000001c0 ae ab e7 f8 50 11 e0 64 41 59 87 d5 10 fe 62 05
000001d0 4c 1f a8 41 f1 84 25 18 de 52 c5 ee 9c 67 49 53
000001e0 b7 e9 b0 75 a1 3d c6 8c 97 2b 72 fb 2e 1a e4 5e
000001f0 59 c5 d4 9c fb 4a 1b bc 59 89 d5 34 fe ba 07 9c
00000200 11 48 67 b1 51 a7 e5 d0 5c e1 ca 44 bd 9b 8d 59
00000210 2f d6 e0 f6 42 35 8c bd 2b 8e f9 26 16 d4 76 f9
00000220 36 16 b4 77 b9 31 96 a5 77 df 30 c2 a2 8f cf 20
00000230 a2 c3 ce 88 a7 33 d2 a8 ef f2 60 2d 40 ef 82 61
00000240 0d 46 2f 94 e1 7a 47 1d 92 4d 6d af 6d e3 6c 4b
00000250 69 bb 75 9b 3d 5a 8f df 20 c2 c2 8e 8f 27 22 d2
00000260 ce ee a6 67 d5 50 ff e2 00 4c 01 a8 05 f0 1c 20
00000270 48 c1 b2 85 af 1d e2 4c 4d a9 ad f5 ec 3c 68 89
00000280 73 37 2a b2 ff ae 01 e4 04 58 19 d0 ff fe 00 04
00000290 00 18 00 50 01 e0 04 40 19 80 13 aa 72 22 0c 67
000002a0 d8 50 51 de e5 46 5a 9d 7f ef 33 27 48 57 00 48
000002b0 01 b0 05 a0 1d c0 4c 81 ab 05 fa 1c 1c 48 49 b1
000002c0 b5 a5 bd dd 8c cd 2a ae ff e6 00 54 01 f8 04 10
000002d0 18 60 ff fe 00 04 00 18 00 50 01 e0 04 40 19 80
000002e0 55 01 fe 04 0c 5a d8 51 51 9e 8d 46 cf 81 d5 5f
000002f0 18 e3 00 1d 02 4b 05 b5 03 a7 15 c9 46 8a a7 08
00000300 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4 20 a5 f3 eb
00000310 0e 5b 01 f8 04 10 18 60 51 41 e7 84 51 19 e6 54
00000320 55 f9 fc 14 08 78 31 10 a6 63 d5 48 ff b2 01 ac
00000330 05 e8 1c 70 49 21 b6 c5 b6 9d b7 4d b3 ad a9 ed
00000340 f4 6c 39 68 97 73 73 2b 2a fa fe 1e 04 44 19 98
00000350 55 51 ff e4 00 58 01 d0 ff fe 00 04 00 18 00 50
00000360 01 e0 04 40 19 80 36 2f d6 67 04 18 18 50 51 e1
00000370 e4 44 59 99 d5 54 ff fa 00 1c 00 48 01 b0 05 a0
00000380 1d c0 4c 81 ab 05 fa 1c 1c 48 49 b1 b5 a5 bd dd
00000390 8c cd 2a ae ff e6 00 54 01 f8 04 10 18 60 51 41
000003a0 e7 84 51 19 e6 54 55 f9 fc 14 08 78 31 10 a6 63
000003b0 d5 48 ff b2 01 ac 05 e8 1c 70 49 21 b6 c5 b6 9d
000003c0 b7 4d b3 ad a9 ed f4 6c 39 68 97 73 73 2b 2a fa
000003d0 fe 1e 04 44 19 98 55 51 ff e4 00 58 01 d0 04 e0
000003e0 1a 40 5d 81 cd 04 ae 1b e4 58 59 d1 d4 e4 fa 5a
000003f0 1d dc 4c c9 aa b5 ff bc 01 88 05 30 1e a0 47 c1
00000400 90 85 63 1f 4a 43 bd 89 8d 35 2e be e7 86 51 15
00000410 e6 7c 55 09 fe 34 04 b8 1b 90 59 61 d7 44 f3 9a
00000420 29 5c f7 ca 30 bc a3 8b c9 38 b6 93 b7 69 b3 75
00000430 ab 3d fa 8c 1f 28 42 f1 8e 25 24 de da c6 de 96
00000440 c7 76 93 37 6a b3 7f ab 01 fa 04 1c 18 48 51 b1
00000450 e5 a4 5d d9 cc d4 aa fb fe 18 04 fd e2 71 3e f5
00000460 0b a0 c6 2f 2d c2 fe 71 77 24 a5 3d 9e ab b9 59
00000470 cd 70 1a b8 08 e9 b0 05 ef c0 36 6c c3 25 96 4a
00000480 65 d2 4e 53 d3 a7 ca 40 5a 4a d0 e2 b5 df df c0
00000490 78 50 a3 d4 93 26 13 46 d3 2b f2 58 3b 23 29 5a
000004a0 4d 78 53 6c 9b 95 4a e7 7c 85 90 00 98 aa d7 91
000004b0 04 86 2c 31 de 6d fb 59 a1 2a 28 d5 ad ee 00 f7

```

```

000004c0 4b 4a a5 88 63 47 d6 75 92 57 8c 42 07 a3 c2 4c
000004d0 63 15 32 26 42 6a cd 05 38 60 bc 90 44 8d 9a 47
000004e0 0f 75 05 74 f2 fb 37 3d e8 d7 c4 e7 58 5e f8 b9
000004f0 37 b2 67 3a df ef 6d 84 7c 02 c6

```

After ASM insertion, frame_len = 1279

```
python3 ./2_datalink_physical_sim_grc/qpsk/tools/analyze_cadu.py ./2_datalink_physical_sim_grc/qpsk/
```

[stage]

file: ./2_datalink_physical_sim_grc/qpsk/data/art/asm_out.bin

size: 9125888 bytes

frame_len=1279, frames=7135.174 (remainder 223)

WARNING: file size is not an integer multiple of frame_len

first 64 bytes:

```

00000000 1a cf fc 1d fb de 00 04 88 18 00 50 68 e2 92 58
00000010 d1 e1 15 01 fe 04 0c 5a d8 50 51 9e 8d 46 cf 81
00000020 d5 5f 32 d6 00 1d 02 4b 05 b5 03 a7 15 c9 46 8a
00000030 a7 08 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4 20 a5

```

ASM occurrences: 7136

ASM @ offsets (first 5): [0, 1279, 2558, 3837, 5116]

bytes around first ASM (-8..+16):

```

00000000 1a cf fc 1d fb de 00 04 88 18 00 50 68 e2 92 58
00000010 d1 e1 15 01 fe 04 0c 5a

```

ASM frame 0 (full 1279 bytes)

```

00000000 1a cf fc 1d fb de 00 04 88 18 00 50 68 e2 92 58
00000010 d1 e1 15 01 fe 04 0c 5a d8 50 51 9e 8d 46 cf 81
00000020 d5 5f 32 d6 00 1d 02 4b 05 b5 03 a7 15 c9 46 8a
00000030 a7 08 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4 20 a5
00000040 f3 eb 0e 5b 01 f8 04 10 18 60 51 41 e7 84 51 19
00000050 e6 54 55 f9 fc 14 08 78 31 10 a6 63 d5 48 ff b2
00000060 01 ac 05 e8 1c 70 49 21 b6 c5 b6 9d b7 4d b3 ad
00000070 a9 ed f4 6c 39 68 97 73 73 2b 2a fa fe 1e 04 44
00000080 19 98 55 51 ff e4 00 58 01 d0 ff fe 00 04 00 18
00000090 00 50 01 e0 04 40 19 80 36 2f d6 67 0e 48 d8 50
000000a0 50 1e fc 37 4e 9b de 4e 50 a9 00 1d 02 4b 05 b5
000000b0 03 a7 15 c9 46 8a a7 08 f4 13 1c 49 4b b2 b1 a0
000000c0 bb da 84 c4 20 a5 f3 eb 0e 5b 11 e9 16 03 0c 75
000000d0 47 56 ff 9d 4b 02 fa 49 4b e6 dc 35 2a 5b 15 35
000000e0 80 44 fd 61 d5 99 2d 81 2b c7 2c 41 7b 12 82 f0
000000f0 80 aa 8f 74 89 96 95 d0 ca 53 39 68 97 73 73 2b
00000100 2a fa fe 1e 04 44 19 98 55 51 ff e4 00 58 01 d0
00000110 04 e0 1a 40 5d 81 cd 04 ae 1b e4 58 59 d1 d4 e4

```

```

00000120 fa 5a 1d dc 4c c9 aa b5 ff bc 01 88 05 30 1e a0
00000130 47 c1 90 85 63 1f 4a 43 bd 89 8d 35 2e be e7 86
00000140 51 15 e6 7c 55 09 fe 34 04 b8 1b 90 59 61 d7 44
00000150 f3 9a 29 5c f7 ca 30 bc a3 8b c9 38 b6 93 b7 69
00000160 b3 75 ab 3d fa 8c 1f 28 42 f1 8e 25 24 de da c6
00000170 de 96 c7 76 93 37 6a b3 7f ab 01 fa 04 1c 18 48
00000180 51 b1 e5 a4 5d d9 cc d4 aa fb fe 18 04 50 19 e0
00000190 54 41 f9 84 15 18 7e 51 05 e6 1c 54 49 f9 b4 15
000001a0 b8 7d 91 0d 66 2f 54 e3 fa 48 1d b0 4d a1 ad c5
000001b0 ec 9c 6b 49 7b b7 19 b2 55 ad fd ec 0c 68 29 70
000001c0 f7 22 32 cc ae ab e7 f8 50 11 e0 64 41 59 87 d5
000001d0 10 fe 62 05 4c 1f a8 41 f1 84 25 18 de 52 c5 ee
000001e0 9c 67 49 53 b7 e9 b0 75 a1 3d c6 8c 97 2b 72 fb
000001f0 2e 1a e4 5e 59 c5 d4 9c fb 4a 1b bc 59 89 d5 34
00000200 fe ba 07 9c 11 48 67 b1 51 a7 e5 d0 5c e1 ca 44
00000210 bd 9b 8d 59 2f d6 e0 f6 42 35 8c bd 2b 8e f9 26
00000220 16 d4 76 f9 36 16 b4 77 b9 31 96 a5 77 df 30 c2
00000230 a2 8f cf 20 a2 c3 ce 88 a7 33 d2 a8 ef f2 60 2d
00000240 40 ef 82 61 0d 46 2f 94 e1 7a 47 1d 92 4d 6d af
00000250 6d e3 6c 4b 69 bb 75 9b 3d 5a 8f df 20 c2 c2 8e
00000260 8f 27 22 d2 ce ee a6 67 d5 50 ff e2 00 4c 01 a8
00000270 05 f0 1c 20 48 c1 b2 85 af 1d e2 4c 4d a9 ad f5
00000280 ec 3c 68 89 73 37 2a b2 ff ae 01 e4 04 58 19 d0
00000290 ff fe 00 04 00 18 00 50 01 e0 04 40 19 80 13 aa
000002a0 72 22 0c 67 d8 50 51 de e5 46 5a 9d 7f ef 33 27
000002b0 48 57 00 48 01 b0 05 a0 1d c0 4c 81 ab 05 fa 1c
000002c0 1c 48 49 b1 b5 a5 bd dd 8c cd 2a ae ff e6 00 54
000002d0 01 f8 04 10 18 60 ff fe 00 04 00 18 00 50 01 e0
000002e0 04 40 19 80 55 01 fe 04 0c 5a d8 51 51 9e 8d 46
000002f0 cf 81 d5 5f 18 e3 00 1d 02 4b 05 b5 03 a7 15 c9
00000300 46 8a a7 08 f4 13 1c 49 4b b2 b1 a0 bb da 84 c4
00000310 20 a5 f3 eb 0e 5b 01 f8 04 10 18 60 51 41 e7 84
00000320 51 19 e6 54 55 f9 fc 14 08 78 31 10 a6 63 d5 48
00000330 ff b2 01 ac 05 e8 1c 70 49 21 b6 c5 b6 9d b7 4d
00000340 b3 ad a9 ed f4 6c 39 68 97 73 73 2b 2a fa fe 1e
00000350 04 44 19 98 55 51 ff e4 00 58 01 d0 ff fe 00 04
00000360 00 18 00 50 01 e0 04 40 19 80 36 2f d6 67 04 18
00000370 18 50 51 e1 e4 44 59 99 d5 54 ff fa 00 1c 00 48
00000380 01 b0 05 a0 1d c0 4c 81 ab 05 fa 1c 1c 48 49 b1
00000390 b5 a5 bd dd 8c cd 2a ae ff e6 00 54 01 f8 04 10
000003a0 18 60 51 41 e7 84 51 19 e6 54 55 f9 fc 14 08 78
000003b0 31 10 a6 63 d5 48 ff b2 01 ac 05 e8 1c 70 49 21
000003c0 b6 c5 b6 9d b7 4d b3 ad a9 ed f4 6c 39 68 97 73
000003d0 73 2b 2a fa fe 1e 04 44 19 98 55 51 ff e4 00 58
000003e0 01 d0 04 e0 1a 40 5d 81 cd 04 ae 1b e4 58 59 d1
000003f0 d4 e4 fa 5a 1d dc 4c c9 aa b5 ff bc 01 88 05 30
00000400 1e a0 47 c1 90 85 63 1f 4a 43 bd 89 8d 35 2e be
00000410 e7 86 51 15 e6 7c 55 09 fe 34 04 b8 1b 90 59 61

```

```

00000420 d7 44 f3 9a 29 5c f7 ca 30 bc a3 8b c9 38 b6 93
00000430 b7 69 b3 75 ab 3d fa 8c 1f 28 42 f1 8e 25 24 de
00000440 da c6 de 96 c7 76 93 37 6a b3 7f ab 01 fa 04 1c
00000450 18 48 51 b1 e5 a4 5d d9 cc d4 aa fb fe 18 04 fd
00000460 e2 71 3e f5 0b a0 c6 2f 2d c2 fe 71 77 24 a5 3d
00000470 9e ab b9 59 cd 70 1a b8 08 e9 b0 05 ef c0 36 6c
00000480 c3 25 96 4a 65 d2 4e 53 d3 a7 ca 40 5a 4a d0 e2
00000490 b5 df df c0 78 50 a3 d4 93 26 13 46 d3 2b f2 58
000004a0 3b 23 29 5a 4d 78 53 6c 9b 95 4a e7 7c 85 90 00
000004b0 98 aa d7 91 04 86 2c 31 de 6d fb 59 a1 2a 28 d5
000004c0 ad ee 00 f7 4b 4a a5 88 63 47 d6 75 92 57 8c 42
000004d0 07 a3 c2 4c 63 15 32 26 42 6a cd 05 38 60 bc 90
000004e0 44 8d 9a 47 0f 75 05 74 f2 fb 37 3d e8 d7 c4 e7
000004f0 58 5e f8 b9 37 b2 67 3a df ef 6d 84 7c 02 c6

```

Post-convolution (packed 2558)

```
python3 ./2_datalink_physical_sim_grc/qpsk/tools/analyze_cadu.py ./2_datalink_physical_sim_grc/qpsk/
```

```
[stage]
```

```
file: ./2_datalink_physical_sim_grc/qpsk/data/art/conv_out.bin
```

```
size: 12288 bytes
```

```
frame_len=2558, frames=4.804 (remainder 2056)
```

```
WARNING: file size is not an integer multiple of frame_len
```

```
first 64 bytes:
```

```

00000000 03 91 85 3e 8f f1 64 ab 01 0b 09 c2 99 c0 00 34
00000010 28 3f ef 9c 5c 00 37 b4 89 16 c6 24 2e 6e b9 a7
00000020 b8 5b 15 9a 8c c8 04 b3 a6 3c 59 f4 fb e7 20 6a
00000030 86 4b 6b b4 87 b7 2f f9 4a c8 6c 72 22 3e b5 9f

```

```
ASM not found.
```

Note: as expected for a non-systematic rate-1/2 encoder, the literal ASM bytes are not present.

CADU frame 0 (full 2558 bytes)

```

00000000 03 91 85 3e 8f f1 64 ab 01 0b 09 c2 99 c0 00 34
00000010 28 3f ef 9c 5c 00 37 b4 89 16 c6 24 2e 6e b9 a7
00000020 b8 5b 15 9a 8c c8 04 b3 a6 3c 59 f4 fb e7 20 6a
00000030 86 4b 6b b4 87 b7 2f f9 4a c8 6c 72 22 3e b5 9f
00000040 ab df 33 ed d8 0c aa a1 75 c0 03 ab e8 bd 0a e6
00000050 9d 47 69 1a c4 be af 98 49 08 e3 77 97 72 16 f2
00000060 0c 28 4a a3 05 52 88 41 c2 d8 93 2b 97 96 40 e0
00000070 93 5e e1 62 1d 40 09 f6 b1 18 1c 23 f6 3e 1e e5
00000080 9d d8 eb 84 5d 9a 1e a9 2d 73 a6 05 9c 34 f8 4f
00000090 b3 9c 65 c5 f7 b7 cb 48 a5 a3 99 a8 cc b7 fc 2f
000000a0 f9 a0 32 40 7c 83 ed 06 3a 31 64 7b 4b d3 d6 59

```

```

000000b0 92 72 3c ff 6e eb 3c 0b 55 1f 30 98 05 8f cb e0
000000c0 4e c3 91 85 d7 37 5e 4a 2f a8 9d 8a 44 2b ae 8d
000000d0 52 14 52 e0 69 14 65 87 35 17 ce 6f 85 23 df 46
000000e0 46 b2 89 7d 71 e2 b2 22 d9 a1 ea a6 fc 99 00 f4
000000f0 3b f4 0c 93 50 90 fb 78 4b 4c 5a 65 99 f4 cf cf
00000100 f8 9f c3 c0 6b 83 33 07 16 3f fc 6d 3b 00 37 67
00000110 5c 03 ab e8 59 8f ff fc 59 c0 00 34 fb 00 03 9c
00000120 5c 00 37 b4 b0 03 a5 99 c0 34 cf fb 03 9f c0 5c
00000130 0e 48 b8 17 42 11 4c 3f 4a 9a 1d e8 08 8b 6b b4
00000140 87 b4 b3 a5 70 41 69 4b ce 61 f9 be c9 c2 ad d1
00000150 1e 74 6e 02 63 b0 03 ab e8 bd 0a e6 9d 47 69 1a
00000160 c4 be af 98 49 08 e3 77 97 72 16 f2 0c 28 4a a3
00000170 05 52 88 41 c2 d8 93 2b 97 96 40 e0 93 5e e1 62
00000180 1d 40 09 f6 b1 18 1c 23 f6 3e 1e e5 9d d8 eb 84
00000190 5d 9a 1e a9 2e 3c 15 49 60 c6 75 ce 71 97 2d bd
000001a0 f0 71 bd fd 9c 4f c6 37 dc 96 9d 7d d7 78 96 eb
000001b0 97 96 77 60 e1 7f a9 46 c9 50 25 a9 2e 08 0a f6
000001c0 17 5c 34 cf 12 82 11 76 6b df d4 c3 1e 6a 4b 5f
000001d0 42 53 ba 5d 87 a9 e3 f8 75 e7 ee 32 5d 21 d7 a6
000001e0 a3 ec de 0c c1 c5 5c ce 58 3c 73 2a a5 54 e0 e8
000001f0 57 c9 e5 7a cf d1 ea a6 fc 99 00 f4 3b f4 0c 93
00000200 50 90 fb 78 4b 4c 5a 65 99 f4 cf cf f8 9f c3 c0
00000210 6b 83 33 07 16 3f fc 6d 3b 00 37 67 5c 03 ab e8
00000220 b0 34 11 29 c3 91 56 3b 37 50 3b 5f a8 43 58 84
00000230 25 38 ea 52 c7 6d 0c 67 6b 64 f7 eb 1b dc a1 1d
00000240 d2 b8 95 aa 61 6b 0c 0f 93 1c f0 c7 7d bc cf 6a
00000250 2d 3f cb 0a 67 03 9c 8f ec 37 ba c1 73 a5 47 12
00000260 f4 c1 6d 64 9f 13 63 db 8d 7b b2 d6 e2 55 e6 35
00000270 ac 09 fb 8c 70 b8 56 f6 c9 64 34 b7 b3 d3 99 a5
00000280 f2 77 fc c8 ee a0 3f ad 50 83 04 60 4a 3c 57 85
00000290 8b 34 26 4e 9f 92 fd d3 87 64 2a 76 6b d2 0e bf
000002a0 12 68 d1 0d 6f 2e 54 e3 4e 5c 1d b9 dc b1 ad 7a
000002b0 b9 dc 62 41 7a b7 ad 12 41 a4 65 6d 1c d7 c3 65
000002c0 be 23 3b cd 1a bf 53 e5 c1 08 71 cb 14 a6 db 9d
000002d0 18 f6 d7 a5 ec b6 24 c9 79 ba df b2 7d 46 85 ee
000002e0 21 72 49 5a 92 ed b0 c3 65 6d cf 3b c3 6b cf 53
000002f0 3b 13 c8 7f 5d 73 a6 08 a2 f4 f8 a8 a4 9c 68 28
00000300 db b7 22 2e d5 ae aa 56 0c 50 0c 38 f4 40 f3 6c
00000310 95 0c fb 7b 04 fc 5a 5c 5c 34 cc b4 b3 9f f9 99
00000320 f7 80 7f f8 a6 06 00 68 f8 78 07 2c 66 60 6e 77
00000330 ff 87 5e a0 06 68 90 80 7f 2b 4a 06 0e 19 88 78
00000340 d9 fe a6 6e f8 10 ff 54 61 4c 08 17 91 f0 a1 27
00000350 58 c8 96 e8 ef ab 35 2d 44 1f 86 61 51 86 7f 90
00000360 5e 7e 07 44 9e 18 69 5b 99 e7 30 d7 f9 ef ce 20
00000370 79 43 da 86 71 32 c2 7e d7 e7 2e 16 21 ee 59 3a
00000380 99 5c f7 43 f0 bc a9 30 c9 38 37 cf b7 63 a3 c5
00000390 ab b4 b3 4c 15 99 f9 f1 0f f8 78 d4 c0 66 6e 1f
000003a0 07 ff 59 8c 60 08 fe f7 80 ac 14 a6 08 71 18 f8

```

```

000003b0 a6 d5 ec 68 f6 09 77 2c b8 b2 ae 79 69 e0 5e 73
000003c0 39 84 9e ff 7e 5b 94 0a 1c d7 10 89 be 2d 4a bd
000003d0 1a 61 81 25 cf 9e 16 cb c7 99 37 93 67 f7 a7 7b
000003e0 e0 a4 ea 51 88 dd 0c 5e ae a4 f4 90 50 dc 9b 44
000003f0 4e bb d9 55 d1 52 f0 0a 50 64 c0 8c 47 df 0a f5
00000400 62 8c 84 83 a2 fa 5a 34 a4 4c cb 98 d5 ff 97 ee
00000410 08 07 21 58 a0 6e 90 e8 87 53 4d 2a 68 79 e6 0f
00000420 26 79 f8 ce fe 78 6f d4 1e 67 42 11 9f e9 29 5f
00000430 81 36 30 86 17 bb ca 79 25 53 8e 76 c0 76 de b7
00000440 06 b6 91 ac 71 b3 5c 76 dd f8 b6 b6 a8 69 b1 b0
00000450 27 3d dd c2 ef 2a ab 25 4e 00 1e c1 d8 01 97 1a
00000460 e0 1f 2d c5 81 8e 6b 4e 1e df 19 d9 96 8d fa ff
00000470 32 e8 44 0f e5 25 50 c1 c6 c0 4f 1b 77 05 cd da
00000480 ac 4b ea c0 75 91 07 06 8f 54 6c 72 c8 17 76 e7
00000490 a1 2a b5 e4 96 01 89 db 38 1e ba df 61 91 46 8b
000004a0 9f 51 72 97 88 52 e3 26 a4 65 be f0 d7 cd 14 ce
000004b0 23 e5 1f da b1 c5 82 c1 db 4e 27 1a d9 da ed c6
000004c0 fa c5 6b 74 47 43 1a 95 69 3d c3 03 37 2b 3c 3f
000004d0 ae 1f 33 04 59 8f fc 54 fe c0 34 1c 17 03 91 b1
000004e0 2c 37 5d d6 73 a8 aa 3e f4 28 0b 14 92 20 9d 1b
000004f0 6a 8b a5 db 02 94 ca dc 23 1f 86 b2 bd 86 71 e1
00000500 2e 7e d9 96 5e 16 ff 3c 99 34 0f 3b f7 90 cf 50
00000510 a7 4f c8 48 e9 c3 a5 ad 3b 34 cc 67 5f 9f f7 e8
00000520 59 8f ff fc 59 c0 00 34 fb 00 03 9c 5c 00 37 b4
00000530 b0 03 a5 99 c0 34 cf fb 03 9f c0 5c 03 41 1f 7c
00000540 28 47 43 f3 fe 27 2e ff ae fb 6b b4 87 b7 1b 02
00000550 73 de 40 72 f2 aa b2 87 d2 d3 fc b0 a8 0f 0c 74
00000560 7e 58 db 8e 3a 70 34 28 ec 03 92 2d 70 37 6a 62
00000570 c3 ab 0f a7 34 1c c4 ef 91 bf 5d 47 5d 08 a1 68
00000580 a4 a8 93 28 d8 2b 7e 2e e2 1a 1a 55 a9 c9 cc 0c
00000590 3b bb f0 f3 55 50 cc f8 00 4f fc 60 05 c0 37 80
000005a0 4b 03 a6 05 9c 34 f8 4f b3 9c 65 c5 29 8f ff fc
000005b0 59 c0 00 34 fb 00 03 9c 5c 00 37 b4 b0 03 a5 99
000005c0 c0 34 cf fb 03 9f c0 5c 37 83 04 b3 a6 3c 59 f4
000005d0 fb e7 20 6a 86 4b 6b b7 c8 07 2f f9 4a c8 6c 72
000005e0 22 3e b5 9f ab df 33 ed d5 ec b6 27 b1 70 03 ab
000005f0 e8 bd 0a e6 9d 47 69 1a c4 be af 98 49 08 e3 77
00000600 97 72 16 f2 0c 28 4a a3 05 52 88 41 c2 d8 93 2b
00000610 97 96 40 e0 93 5e e1 62 1d 40 09 f6 b1 18 1c 23
00000620 f6 3e 1e e5 9d d8 eb 84 5d 9a 1e a9 2d 73 a6 05
00000630 9c 34 f8 4f b3 9c 65 c5 f7 b7 cb 48 a5 a3 99 a8
00000640 cc b7 fc 2f f9 a0 32 40 7c 83 ed 06 3a 31 64 7b
00000650 4b d3 d6 59 92 72 3c ff 6e eb 3c 0b 55 1f 30 98
00000660 05 8f cb e0 4e c3 91 85 d7 37 5e 4a 2f a8 9d 8a
00000670 44 2b ae 8d 52 14 52 e0 69 14 65 87 35 17 ce 6f
00000680 85 23 df 46 46 b2 89 7d 71 e2 b2 22 d9 a1 ea a6
00000690 fc 99 00 f4 3b f4 0c 93 50 90 fb 78 4b 4c 5a 65
000006a0 99 f4 cf cf f8 9f c3 c0 6b 83 33 07 16 3f fc 6d

```

```

000006b0 3b 00 37 67 5c 03 ab e8 59 8f ff fc 59 c0 00 34
000006c0 fb 00 03 9c 5c 00 37 b4 b0 03 a5 99 c0 34 cf fb
000006d0 03 9f c0 5c 0e 48 b8 17 42 11 4c 3f 4a 44 f8 9c
000006e0 5f 9c 6b b4 87 b7 15 9a 65 ad 0f cf cc 64 c3 c3
000006f0 f7 df 33 30 a2 8f ff c8 a2 c0 03 a8 a7 00 34 28
00000700 ec 03 92 2d 70 37 6a 62 c3 ab 0f a7 34 1c c4 ef
00000710 91 bf 5d 47 5d 08 a1 68 a4 a8 93 28 d8 2b 7e 2e
00000720 e2 1a 1a 55 a9 c9 cc 0c 3b bb f0 f3 55 50 cc f8
00000730 00 4f fc 60 05 c0 37 80 4b 03 a6 05 9c 34 f8 4f
00000740 b3 9c 65 c5 f7 b7 cb 48 a5 a3 99 a8 cc b7 fc 2f
00000750 f9 a0 32 40 7c 83 ed 06 3a 31 64 7b 4b d3 d6 59
00000760 92 72 3c ff 6e eb 3c 0b 55 1f 30 98 05 8f cb e0
00000770 4e c3 91 85 d7 37 5e 4a 2f a8 9d 8a 44 2b ae 8d
00000780 52 14 52 e0 69 14 65 87 35 17 ce 6f 85 23 df 46
00000790 46 b2 89 7d 71 e2 b2 22 d9 a1 ea a6 fc 99 00 f4
000007a0 3b f4 0c 93 50 90 fb 78 4b 4c 5a 65 99 f4 cf cf
000007b0 f8 9f c3 c0 6b 83 33 07 16 3f fc 6d 3b 00 37 67
000007c0 5c 03 ab e8 b0 34 11 29 c3 91 56 3b 37 50 3b 5f
000007d0 a8 43 58 84 25 38 ea 52 c7 6d 0c 67 6b 64 f7 eb
000007e0 1b dc a1 1d d2 b8 95 aa 61 6b 0c 0f 93 1c f0 c7
000007f0 7d bc cf 6a 2d 3f cb 0a 67 03 9c 8f ec 37 ba c1
00000800 73 a5 47 12 f4 c1 6d 64 9f 13 63 db 8d 7b b2 d6
00000810 e2 55 e6 35 ac 09 fb 8c 70 b8 56 f6 c9 64 34 b7
00000820 b3 d3 99 a5 f2 77 fc c8 ee a0 3f ad 50 83 04 60
00000830 4a 3c 57 85 8b 34 26 4e 9f 92 fd d3 87 64 2a 76
00000840 6b d2 0e bf 12 68 d1 0d 6f 2e 54 e3 4e 5c 1d b9
00000850 dc b1 ad 7a b9 dc 62 41 7a b7 ad 12 41 a4 65 6d
00000860 1c d7 c3 65 be 23 3b cd 1a bf 53 e5 c1 08 71 cb
00000870 14 a6 db 9d 18 f6 d7 a5 ec b6 24 c9 79 ba df b2
00000880 7d 46 85 ee 21 72 49 5a 92 ed b0 c3 65 6d cf 3b
00000890 c3 6b cf 53 3b 13 c8 7f 5d 73 a6 08 a2 f4 f8 a8
000008a0 a4 9c 68 28 db b7 22 2e d5 ae aa 56 0c 50 0c 38
000008b0 f4 40 f3 6c 95 0c fb 7b 04 fc 5a 5c 5c 34 12 82
000008c0 c2 94 c4 49 31 2b b0 91 c4 6d 43 a2 27 2e c8 17
000008d0 ab aa 7f aa d7 4c 63 49 05 00 f7 7a e5 e5 7a 25
000008e0 f8 f9 47 cf 80 3d e4 d4 f4 43 62 3a 73 91 bf 8e
000008f0 9c d3 06 f9 be 2d 70 37 5e 70 41 67 0e 48 8c e2
00000900 30 19 7c 79 a4 2a 41 e5 eb 32 90 e5 ba d1 1e 7a
00000910 25 26 1f 98 ad b9 e6 3b 37 6a 56 e5 36 98 5a 24
00000920 23 6a 20 01 32 c1 31 67 3a 59 ab b4 6e dc 55 1c
00000930 9b ad cc 77 06 81 f5 b2 21 96 cc 93 b4 1b 79 a7
00000940 52 ac e0 40 bc 9e 54 da 56 df 62 e9 ab ba f8 52
00000950 07 7e fd e4 30 95 38 d3 70 1d b4 db 64 13 b0 00
00000960 d0 70 82 0c f6 a2 ea d0 ff 84 28 d5 c8 19 d9 72
00000970 db 02 a0 e1 71 0b da 14 cd 61 82 50 1f 2d c8 6f
00000980 da 86 72 73 e9 c0 e9 5c ce 56 a9 55 0c 25 a7 8f
00000990 d5 cb 85 b1 6e 11 4f 7d 14 1e b9 4e e9 7b 23 f6
000009a0 3e fa 54 ac 56 6a 0a dc 2e cb b2 08 0a cc 43 c7

```

```

000009b0 31 36 07 db f5 33 58 87 ba 12 a5 c5 1d 7a b7 a3
000009c0 84 cf 28 08 88 cd 56 01 8a 99 5c cd c4 87 8e 0e
000009d0 62 6b a7 7b e3 3b f4 e5 c2 4a c8 62 dd 53 11 13
000009e0 7d 17 6b 5e 70 75 41 4d dd fb 27 e0 77 3f 44 df
000009f0 46 71 3c b0 9f 51 4b 68 c1 6d 67 0d d9 ee

```

Summary Table

Stage	Expected len	Analyzer len	ASM found?	Notes
TM (pre-ASM)	1115	1115	No	Frames: 38 exact
Randomized TM	1115	1115	No	Fractional tail (stop mid-frame)
RS/interleave	1275	1275	No	Fractional tail (stop mid-frame)
ASM inserted	1279	1279	Yes	33 occurrences; aligned
Post-conv (packed)	2558	2558	No	Looks correct

Repro commands

All commands executed from the qpsk directory.

```
python3 ./grc/sync_coding/sync_coding_v0.py
```

```
# COMPARE FRAMES + LOOK FOR ASM
```

```
python3 ./tools/analyze_cadu.py ./data/art/tm_out.bin 1115
```

```
python3 ./tools/analyze_cadu.py ./data/art/rand_out.bin 1115
```

```
python3 ./tools/analyze_cadu.py ./data/art/rs_out.bin 1275
```

```
python3 ./tools/analyze_cadu.py ./data/art/asm_out.bin 1279
```

```
python3 ./tools/analyze_cadu.py ./data/art/conv_out.bin 2558
```

```
# VIEW HEX FRAME
```

```
python3 ./tools/dump_frame.py ./data/SPPencaps_CFDp_metadata_tx.bin 1115 > ./data/pkt.hex
```

```
python3 ./tools/dump_frame.py ./data/art/tag_out.bin 1115 > ./data/art/tag_frame0.hex
```

```
python3 ./tools/dump_frame.py ./data/art/tm_out.bin 1115 > ./data/art/tm_frame0.hex
```

```
python3 ./tools/dump_frame.py ./data/art/rs_out.bin 1275 > ./data/art/rs_frame0.hex
```

```
python3 ./tools/dump_frame.py ./data/art/asm_out.bin 1279 > ./data/art/asm_frame0.hex
```

```
# CHECK THE HOST PORT
```

```
ss -ulpn | grep 52001
```

```
ss -ulpn | grep 52002
```

```
ss -ulpn | grep 52003
```